THE V AND C CONDITIONS FOR THE COMPARISON OF THE ETALE AND FPPF TOPOLOGIES, WITH THE FIVE COMPARISON STATEMENTS THEY YIELD

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is the author's complete argument: each of the five parts is reduced to the corresponding general statement, and part (5) is carried out by the four-term chain of equalities plus the explicit induction on i showing that R epsilon_{Y,*} reflects isomorphisms for maps out of a_Y^{-1}L.

Editorial note. The label lemma-V-C-all-n-etale-fppf names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The statement and proof are copied from the pinned chapter file; the eight cited inputs are kept as external pointers with their canonical Stacks tags.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Comparison morphisms a_X, pi_X, epsilon_X between the etale and fppf topoi over S. Same chapter, the morphisms of topoi relating X_etale, (Sch/X)_fppf and (Sch/X)_etale and the composition a_X = epsilon_X composed with the comparison. Supplies every functor appearing in the five assertions.
- The general V and C conditions of the sites chapter. Cohomology on Sites, the general lemmas proving the V-C conditions, the V-implies-C statement and the category A_X of the comparison section. Supplies the abstract statements that the proof specialises to the etale-fppf setting.
- Proper base change for torsion coefficients and comparison of higher direct images. Same chapter, the proper base change theorem and the comparison of the small and big higher direct images for proper morphisms. Supplies the third and fourth equalities in the chain for part (5).

Source reference. Original lemma 075R: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21666–21666. Prerequisite pointer invoked at proof line(s) 21749; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DDJ: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21323–21323. Prerequisite pointer invoked at proof

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line(s) 21794; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0DDM: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 21484–21484. Prerequisite pointer invoked at proof line(s) 21790; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0EZ9: sources/stacks-chow/tags.txt, source lines 18719–18719. Prerequisite pointer invoked at proof line(s) 21753; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0EZF: sources/stacks-chow/tags.txt, source lines 18725–18725. Prerequisite pointer invoked at proof line(s) 21759; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0EZA: sources/stacks-chow/tags.txt, source lines 18720–18720. Prerequisite pointer invoked at proof line(s) 21763; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0D6E: sources/stacks-chow/tags.txt, source lines 16519–16519. Prerequisite pointer invoked at proof line(s) 21792; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original section 0EZ1: sources/stacks-chow/tags.txt, source lines 18711–18711. Prerequisite pointer invoked at proof line(s) 21750; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *With notation as above.*

(1) *For $X \in \mathrm{Ob}((Sch/S)_{fppf})$ and an abelian sheaf $\mathcal{F}$ on $X_{étale}$ we have $\epsilon_{X,*}a_X^{-1}\mathcal{F} = \pi_X^{-1}\mathcal{F}$ and $R^i\epsilon_{X,*}(a_X^{-1}\mathcal{F}) = 0$ for $i > 0$.*
(2) *For a finite morphism $f : X \to Y$ in $(Sch/S)_{fppf}$ and abelian sheaf $\mathcal{F}$ on X we have $a_Y^{-1}(R^i f_{small,*}\mathcal{F}) = R^i f_{big,fppf,*}(a_X^{-1}\mathcal{F})$ for all i.*
(3) *For a scheme X and K in $D^+(X_{étale})$ the map $\pi_X^{-1}K \to R\epsilon_{X,*}(a_X^{-1}K)$ is an isomorphism.*
(4) *For a finite morphism $f : X \to Y$ of schemes and K in $D^+(X_{étale})$ we have $a_Y^{-1}(Rf_{small,*}K) = Rf_{big,fppf,*}(a_X^{-1}K)$.*
(5) *For a proper morphism $f : X \to Y$ of schemes and K in $D^+(X_{étale})$ with torsion cohomology sheaves we have $a_Y^{-1}(Rf_{small,*}K) = Rf_{big,fppf,*}(a_X^{-1}K)$.*

Proof. By Lemma 075R the lemmas in Cohomology on Sites, Section 0EZ1 all apply to our current setting. To translate the results observe that the category $\mathcal{A}_X$ of Cohomology on Sites, Lemma 0EZ9 is the essential image of $a_X^{-1} : Ab(X_{étale}) \to Ab((Sch/X)_{fppf})$.

Part (1) is equivalent to (V_n) for all n which holds by Cohomology on Sites, Lemma 0EZF.

Part (2) follows by applying ϵ_Y^{-1} to the conclusion of Cohomology on Sites, Lemma 0EZA.

Part (3) follows from Cohomology on Sites, Lemma 0EZF part (1) because $\pi_X^{-1}K$ is in $D^+_{\mathcal{A}'_X}((Sch/X)_{étale})$ and $a_X^{-1} = \epsilon_X^{-1} \circ a_X^{-1}$.

Part (4) follows from Cohomology on Sites, Lemma 0EZF part (2) for the same reason.

Part (5). We use that

$$\begin{aligned} R\epsilon_{Y,*}Rf_{big,fppf,*}a_X^{-1}K &= Rf_{big,étale,*}R\epsilon_{X,*}a_X^{-1}K \\ &= Rf_{big,étale,*}\pi_X^{-1}K \\ &= \pi_Y^{-1}Rf_{small,*}K \\ &= R\epsilon_{Y,*}a_Y^{-1}Rf_{small,*}K \end{aligned}$$

The first equality by the commutative diagram in Lemma 0DDM and Cohomology on Sites, Lemma 0D6E. The second equality is (3). The third is Lemma 0DDJ part (2). The fourth is (3) again. Thus the base change map $a_Y^{-1}(Rf_{small,*}K) \to Rf_{big,fppf,*}(a_X^{-1}K)$ induces an isomorphism

$$R\epsilon_{Y,*}a_Y^{-1}Rf_{small,*}K \to R\epsilon_{Y,*}Rf_{big,fppf,*}a_X^{-1}K$$

The proof is finished by the following remark: a map $\alpha : a_Y^{-1}L \to M$ with L in $D^+(Y_{étale})$ and M in $D^+((Sch/Y)_{fppf})$ such that $R\epsilon_{Y,*}\alpha$ is an isomorphism, is an isomorphism. Namely, we show by induction on i that $H^i(\alpha)$ is an isomorphism. This is true for all sufficiently small i. If it holds for $i \leq i_0$, then we see that $R^j\epsilon_{Y,*}H^i(M) = 0$ for $j > 0$ and $i \leq i_0$ by (1) because $H^i(M) = a_Y^{-1}H^i(L)$ in this range. Hence $\epsilon_{Y,*}H^{i_0+1}(M) = H^{i_0+1}(R\epsilon_{Y,*}M)$ by a spectral sequence argument. Thus $\epsilon_{Y,*}H^{i_0+1}(M) = \pi_Y^{-1}H^{i_0+1}(L) = \epsilon_{Y,*}a_Y^{-1}H^{i_0+1}(L)$. This implies $H^{i_0+1}(\alpha)$ is an isomorphism (because $\epsilon_{Y,*}$ reflects isomorphisms as it is the identity on underlying presheaves) as desired. □